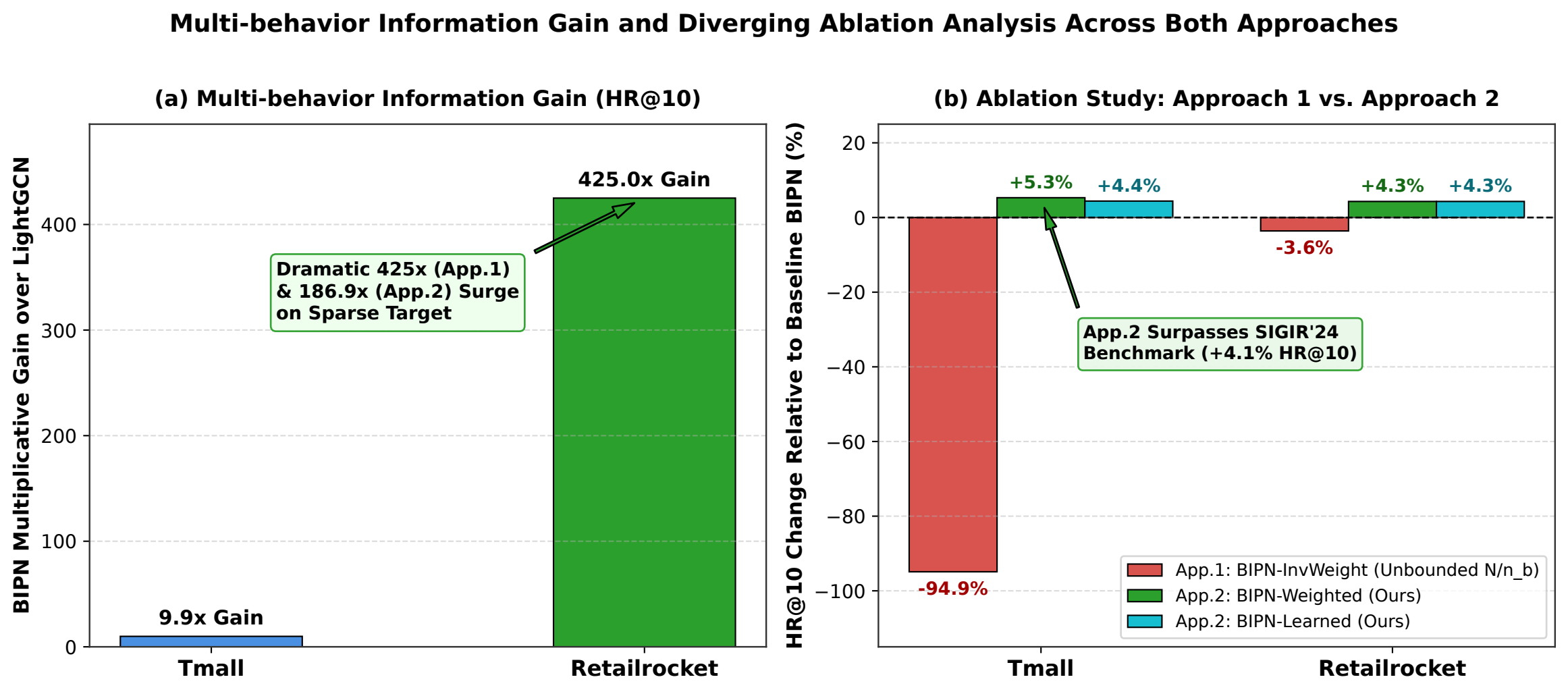
Multi-behavior Information Gain and Diverging Ablation Analysis Across Both Approaches
(a) Multi-behavior Information Gain (HR@10)
BIPN Multiplicative Gain over LightGCN
0
100
200
300
400
9.9x Gain
425.0x Gain
Dramatic 425x (App.1)
& 186.9x (App.2) Surge
on Sparse Target
Tmall
Retailrocket
(b) Ablation Study: Approach 1 vs. Approach 2
HR@10 Change Relative to Baseline BIPN (%)
20
0
−20
−40
−60
−80
−100
+5.3%
+4.4%
+4.3%
+4.3%
-3.6%
-94.9%
App.2 Surpasses SIGIR'24
Benchmark (+4.1% HR@10)
App.1: BIPN-InvWeight (Unbounded N/n_b)
App.2: BIPN-Weighted (Ours)
App.2: BIPN-Learned (Ours)
Tmall
Retailrocket